STAT 527
Robustness and Quantile Regression

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References

- Huber 1984, "Finite sample breakdown of M and P estimators."
- Koenker, Bassett, 1978, "Regression Quantiles"

M-estimator

Given sample $X = \{x_1, x_2, \ldots, x_n\}$, define

$$T(X) := \arg\min_T \sum_{i=1}^n \rho(x_i - T) \tag{1}$$

This can be interpreted as MLE for linear regression with design matrix $(1, \ldots, 1)^\top$ and i.i.d. noise with distribution $\exp(-\rho(x))$.

Examples

- $\rho(x) = x^2$
- $\rho(x) = |x|$

$T(\lambda x) = \lambda T(x)$

A scale invariant version

$$T := \arg\min_T \sum_{i=1}^{n} \rho\left(\frac{x_i - T}{MAD}\right) \tag{2}$$

$= \text{median}\{|x_i - \text{med}(x)|\}$

where MAD is the median absolute deviation from the median. We will see that both (1) and (2) have breakdown point equal to 1/2 (assuming ρ' is monotone, bounded, and $\rho'(-\infty) = -\rho'(\infty)$).

P-estimator

a probability measure

$$T(X) := \frac{\int \exp(-\sum \rho(x_i - \theta))\theta d\theta}{\int \exp(-\sum \rho(x_i - \theta))d\theta} \tag{3}$$

Finite sample breakdown

ϵ-contamination

multiset

Sample: $X = (x_1, \ldots, x_n)$; adjoin $Y = (y_1, \ldots, y_m)$ so that the corrupted sample

$$X' = X \cup Y,$$

where $\epsilon = \frac{m}{n+m}$.

ϵ-replacement

X' is obtained by replacing a subset of size m of the sample X by Y, where $\epsilon := \frac{m}{n}$.

Maximum bias

$$b(\epsilon; X, T) := \sup |T(X') - T(X)| \tag{4}$$

where the supremum is over all ϵ-corrupted sample X'.

Breakdown point

$$\epsilon^*(X, T) = \inf\{\epsilon \,|b(\epsilon; X, T) = \infty\}. \tag{5}$$

$0 \leq \epsilon \leq 1$

Examples:

- breakdown point can be as high as 1 (T is constant).
- Sample mean has breakdown point at $\frac{1}{n+1}$. (ϵ-contamination model)
- Sample median breaks down at $\epsilon^* = \frac{1}{2}$.
- $\epsilon^* = \frac{1}{2}$ is the highest breakdown point achievable by translation equivariant estimator.

Proof that $\varepsilon^*(X,T) \leq \frac{1}{2}$ for translation equivariant T:

Translation equivariance means:

$$T(X+\lambda) = \lambda + T(X), \quad \forall\, \lambda \in \mathbb{R}^+$$

From the definition of b we have:

$$|T(X \cup (X+\lambda)) - T(X)| \leq b(\tfrac{1}{2}, X, T)$$

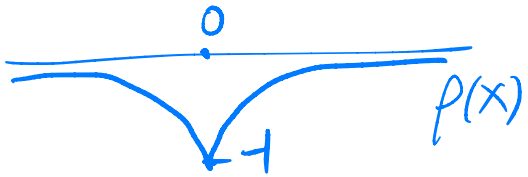
$$|T(X \cup (X+\lambda)) - T(X+\lambda)| \leq b(\tfrac{1}{2}, X+\lambda, T)$$
$$= b(\tfrac{1}{2}, X, T)$$

Also, $|T(X) - T(X+\lambda)| = \lambda$. By the triangle ineq.,

$$b(\tfrac{1}{2}, X, T) \geq \lambda/2$$

Since λ is arbitrary, $b(\tfrac{1}{2}, X, T) = \infty$

Theorem for bounded ρ



Assume that ρ has a minimum at 0, $\rho(0) = -1$, and ρ increases monotonically toward both sides with $\lim_{|x|\to\infty} \rho(x) = 0$. Let

$$A = -\sum_X \rho(x_i - T(X)), \tag{6}$$

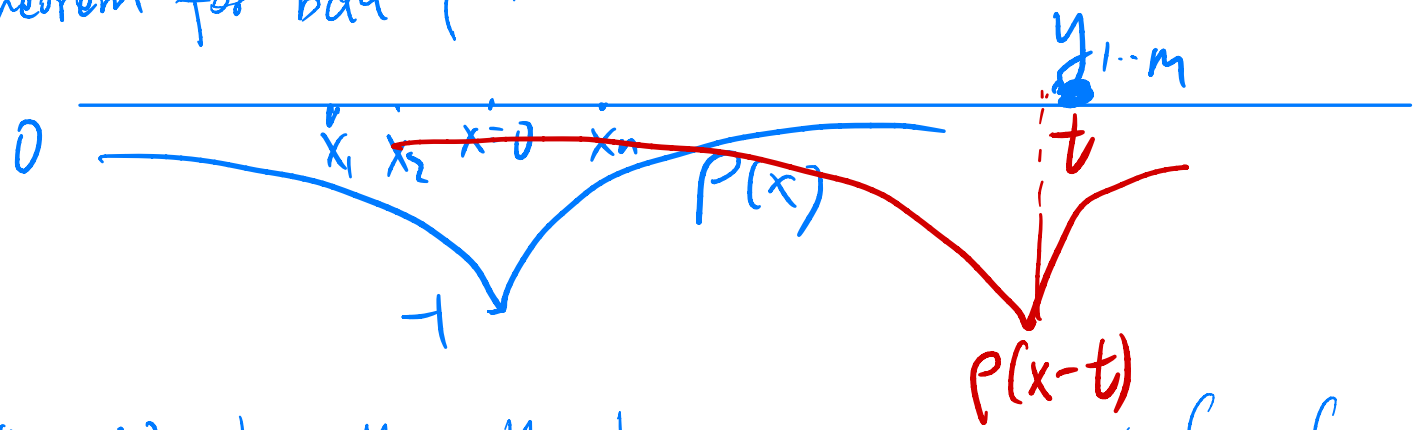
then the ϵ-contamination breakdown point of T is

$$\frac{m^*}{n + m^*},$$

where m^* is an integer satisfying $\lceil A \rceil \leq m^* \leq \lfloor A \rfloor + 1$.

Proof sketch: See next page

Idea of theorem for bdd ρ:



$T(X) = \arg\min_T \{\sum_x \rho(x_i - T)\}$. Let $y_1 = \cdots = y_m$ be contamination points from far away.

When is $T(X')$ close to $y_1 = \cdots = y_m$ rather than $T(X)$?

- If $t = T(X) \Rightarrow \sum_{x'} \rho(x_i' - t) \approx -A$
- If $t = y_1 = \cdots = y_m$

$$\Rightarrow \sum_{x'} \rho(x_i' - t) = \sum_{x} \rho(\ ) + \sum_{y} \rho(\ )$$
$$\approx 0 + (-m)$$
$$= -m$$

Thus $T(X') \approx y_1 = \cdots = y_m$ roughly when $m > A$

Ref: Koenker – Bassett 1978

Background and motivation for quantile regression

Interpretation of M-estimator as MLE

Noise $\frac{1}{z}\exp(-\rho(x))$

$$T(X) := \arg\min_T \sum_{i=1}^{n} \rho(x_i - T)$$

$x_i = \theta + \text{noise}$

The dogma of normality

Poincaré (quoted by Cramér): "everyone believes in the [Gaussian] law of errors, the experimenters because they think it is a mathematical theorem, the mathematicians because they think it is an experimental fact."

Beyond least squares

$T = \operatorname{argmin}_t \sum_{i=1}^{n} |t - x_i|$

In 1818 Laplace proved that in bivariate regression through the origin, the least absolute error (LAE) estimator had smaller asymptotic variance than the least squares estimator if the variance σ^2 and the density f of the error satisfied $[2f(0)]^{-1} < \sigma$.

$\frac{1/I(\theta)}{\mathrm{Var}(T(X))}$

"Quick and dirty" estimators

Gastwirth $0.3F_n^{-1}(\frac{1}{3}) + 0.4F_n^{-1}(\frac{1}{2}) + 0.3F_n^{-1}(\frac{2}{3})$. Achieves asymptotic efficiency of nearly 80% for the Gaussian, Laplace, logistic and Cauchy distributions.

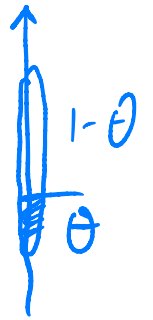
Quantile regression

$$-\sum_{t:\,y_t \geq b} \theta + \sum_{t:\,y_t < b} (1-\theta) = 0$$

$$\Rightarrow \quad \theta \,|\{t : y_t \geq b\}| = (1-\theta)\,|\{t : y_t < b\}|$$

θth sample quantile

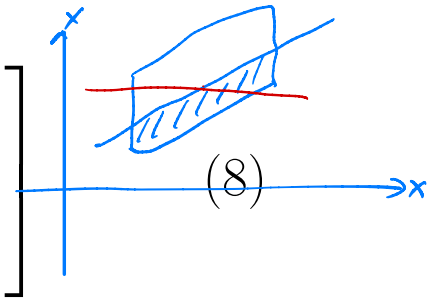
$$\arg\min_{b \in \mathbb{R}} \left[\sum_{t:\, y_t \geq b} \theta |y_t - b| + \sum_{t:\, y_t < b} (1-\theta)|y_t - b| \right] \tag{7}$$



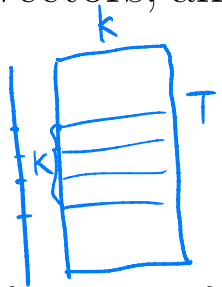
θth regression quantile

Goal: find the θth quantile of $P_{Y|X=x_t}$

$$\arg\min_{b \in \mathbb{R}^K} \left[\sum_{t:\, y_t \geq x_t^\top b} \theta |y_t - x_t^\top b| + \sum_{t:\, y_t < x_t^\top b} (1-\theta)|y_t - x_t^\top b| \right] \tag{8}$$



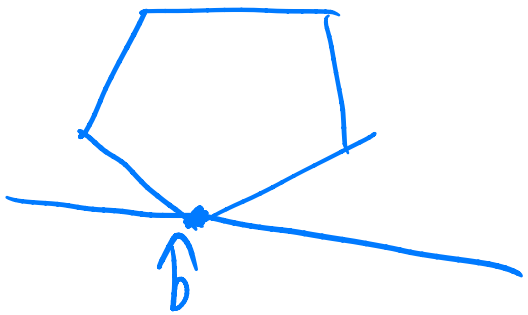
where $\{x_t : t = 1, \ldots, T\}$ is a sequence of row K-vectors, and $\{y_t : t = 1, \ldots, T\}$ is a random sequence of responses.



Characterization of solution

[Thm 3.1, Koenker-Bassett] There exists $h \subseteq \{1, \ldots, T\}$, $|h| = K$, s.t. one solution has the form $\beta^* = X(h)^{-1} y(h)$.

Proof: use the fact that linear programming is solved at a vertex of the constraint polytope.



Background: linear optimization

$\min_b \; c^T b$ where $c \in \mathbb{R}^k$

s.t. $x_t^T b \le y_t$, $t = 1, \ldots, T$

optimizer is a vertex (at least k constraints are satisfied)

In quantile regression, an optimizer $\hat{b}$ for (8) is a local optimizer, so $\hat{b}$ is optimal for the linear optimization problem:

$$\min_b \sum_{t=1}^{T} C_t b$$

$$\text{s.t.} \quad X_t \geq x_t^T b \;, \; t : y_t \geq x_t^T \hat{b}$$

$$y_t < x_t^T b \;, \; t : y_t < x_t^T \hat{b}$$

where $$C_t = \begin{cases} -\theta x_t \;, & \text{if } y_t \geq x_t^T \hat{b} \\ (1-\theta) x_t \;, & \text{if } y_t < x_t^T \hat{b} \end{cases}$$

At least k equalities are achieved in the constraints.

Second proof: (8) is equivalent to the following linear program

$$\min_{c \in \mathbb{R}^T,\, b \in \mathbb{R}^k} \sum_{t=1}^{T} c_t$$

$$\text{s.t.} \quad \left\{ \begin{array}{l} c_t \geq \theta (y_t - x_t^T b) \\ c_t \geq -(1-\theta)(y_t - x_t^T b) \end{array} \right\}, \quad \forall\, t = 1, 2, \cdots T$$

of variables $= T + k$

$\Rightarrow$ at least $T+k$ constraints achieve equality

$\Rightarrow$ $y_t = x_t^T b$ for at least k choices of t.

Asymptotic properties of $\beta^*(\theta)$

- consistency: $\lim_{T\to\infty} \beta^*(\theta) = \beta$?
- asymptotic normality: $\sqrt{T}(\beta^*(\theta) - \mathbb{E}[\beta^*]) \to$?

Local model (special case of one quantile func.)

[Theorem 4.1 in K-B] F cdf with continuous and positive density.

- $\xi(\theta) = F^{-1}(\theta)$, $\theta \in (0,1)$. (want to estimate)
- $\{Y_i\}_{i=1}^T$ samples from F.
- $\xi^*(\theta) = F_T^{-1}(\theta)$, where $F_T = \frac{1}{T}\sum_{i=1}^T 1\{Y_i \le t\}$.

Then $\sqrt{T}(\xi^*(\theta) - \xi(\theta)) \to \mathcal{N}(0, \Omega(0, F))$, where

$$\Omega(\theta, F) = \frac{\theta(1-\theta)}{|f(\xi(\theta))|^2} \tag{9}$$

THEOREM 4.1: *Let* $\{\xi_T^*(\theta_1), \ldots, \xi_T^*(\theta_M)\}$ *with* $0 < \theta_1 < \theta_2 < \ldots < \theta_M < 1$, *denote a sequence of unique sample quantiles from random samples of size* T *from a population with inverse distribution function* $\xi(\theta) = F^{-1}(\theta)$. *If* F *is continuous and has continuous and positive density,* f, *at* $\xi(\theta_i)$, $i = 1, \ldots, M$, *then,*

$$\sqrt{T}[\xi_T^*(\theta_1) - \xi(\theta_1), \ldots, \xi_T^*(\theta_M) - \xi(\theta_M)]$$

converges in distribution to an (M)*-variate Gaussian random vector with mean,* 0, *and covariance matrix* $\Omega(\theta_1, \ldots, \theta_M; F)$ *with typical element,*

$$\omega_{ij} = \frac{\theta_i(1-\theta_j)}{f(\xi(\theta_i))f(\xi(\theta_j))}, \qquad i \leq j.$$

This theorem provides the foundation for a large-sample theory of so-called "systematic statistics"—estimators which are linear combinations of a few sample quantiles.

Regression with an intercept

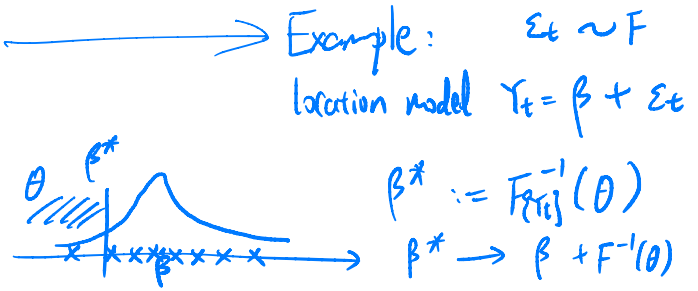
[Thm 4.2 B-K] Given F, $\xi(\theta) := F^{-1}(\theta)$, $0 < \theta < 1$, $\vec{\xi}(\theta) = (\xi(\theta), 0, \ldots, 0) \in \mathbb{R}^K$.

- $x_1, \ldots, x_T \in \mathbb{R}^K$, $Y_1, \ldots, Y_T \in \mathbb{R}$, $\mathbb{P}[Y_t < y] = F[y - x_t^\top \beta]$,
- $\beta_T^*(\theta)$: solution to quantile regression,
- $\xi_T^*(\theta) = \beta_T^*(\theta) - \beta \in \mathbb{R}^K$,
- $x_{1t} = 1$, $t = 1, \ldots,$ $Q := \lim_{T\to\infty} \frac{1}{T} X^T X$ p.s.d.

Then $\sqrt{T}(\xi_T^*(\theta) - \vec{\xi}(\theta)) \to \mathcal{N}(0, \Omega(\theta, F)Q^{-1})$, $T \to \infty$.

Remark

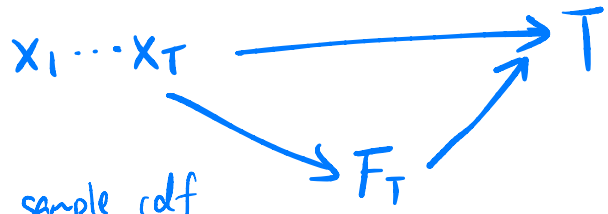
- $\beta_T^*(\theta)$ not consistent unless $\xi(\theta) = 0$,
- If F is symmetric, then $\xi(\frac{1}{2}) = 0$.



Estimator as a function of the cdf

Examples

- $T(x_1, \ldots, x_T) = x_1$?
- $T(x_1, \ldots, x_T) = F_T^{-1}(\frac{1}{2})$.



estimator $= T(F_T)$

Delta method

Influence function ("functional derivative"):

$$I_F(x) := \lim_{\epsilon \to 0} \frac{T((1-\epsilon)F + \epsilon\delta_x) - T(F)}{\epsilon} \tag{10}$$

perturbation direction G

$\Rightarrow \lim_{\varepsilon \to 0} \frac{T((1-\varepsilon)F + \varepsilon G) - T(F)}{\varepsilon} = \int I_F(x) G(x)\, dx$

e.g. $T(x_1, \ldots, x_T) = F_T^{-1}(\frac{1}{2})$

$$\Rightarrow I_F(x) = \begin{cases} \frac{1}{2f(\xi(\frac{1}{2}))} & x < \xi(\frac{1}{2}) \\ -\frac{1}{2f(\xi(\frac{1}{2}))} & x \geq \xi(\frac{1}{2}) \end{cases} \tag{11}$$

Proof of Thm 4.1: The empirical distribution F_T is close to F; we apply the linearization

$$T(F_T) \approx T(F) + \int I_F(x)(dF_T - dF) \tag{12}$$

Then

$$\text{var}(T(F_T)) = \text{var}(\int I_F(x)(dF_T - dF)) \tag{13}$$

$$= \text{var}(\int I_F(x)F_T) \tag{14}$$

Note that $\mathbb{E}[\int I_F(x)dF_T] = \int I_F(x)dF = 0$. Therefore

$$(14) = \mathbb{E}[(\int I_F(x)dF_T)^2] \tag{15}$$

$$= \mathbb{E}[\frac{1}{T}\sum_{i=1}^{T} I_F(x_i)^2] \tag{16}$$

$$= \frac{1}{T}\mathbb{E}[I_F(X)^2] \tag{17}$$

$$= \frac{\theta(1-\theta)}{Tf^2(\xi(\theta))}. \tag{18}$$

We now give another proof of Thm 4.1 which, though seemingly more complicated, can be extended to the regression case (thm 4.2).

Choose a small interval $[a, a+da]$

$$\mathbb{P}\left[\sqrt{T}(T(Y^T) - F^{-1}(\theta)) \in [a, a+da]\right]$$

$$= \sum_{h=1}^{T} \mathbb{P}\left[\sqrt{T}(Y_h - F^{-1}(\theta)) \in [a, a+da]\right] \cdot \mathbb{P}\left[\theta T \text{ samples} < \frac{a}{\sqrt{T}} + F^{-1}(\theta)\right]$$

$$= T\,\mathbb{P}\left[\sqrt{T}(Y_1 - F^{-1}(\theta)) \in [a, a+da]\right] \cdot \mathbb{P}\left[\theta T \text{ samples} < \frac{a}{\sqrt{T}} + F^{-1}(\theta)\right]$$

$$= T \cdot \frac{da}{\sqrt{T}} \cdot f\left(F^{-1}(\theta) + \frac{a}{\sqrt{T}}\right) \binom{T}{\theta T} F\left(F^{-1}(\theta) + \frac{a}{\sqrt{T}}\right)^{\theta T} \left(1 - F\left(F^{-1}(\theta) + \frac{a}{\sqrt{T}}\right)\right)^{(1-\theta)T-1}$$

$$= \underbrace{\left(1 + O\left(\frac{1}{\sqrt{T}}\right)\right) \sqrt{T}\, da\, f(F^{-1}(\theta)) \binom{T}{\theta T}}_{\text{Const. independent of } a} \left(\theta + \frac{a}{\sqrt{T}} f(\xi(\theta)) + \frac{a^2}{2T} f'(\xi(\theta)) + O\left(\frac{1}{T^{3/2}}\right)\right)^{\theta T}$$

$$\left(1 - \theta - \frac{a}{\sqrt{T}} f(\xi(\theta)) - \frac{a^2}{2T} f'(\xi(\theta)) + O\left(\frac{1}{T^{3/2}}\right)\right)^{(1-\theta)T-1}$$

$$\approx \text{const.} \left[\exp\left(\frac{a}{\theta\sqrt{T}} f(\xi(\theta)) + \frac{a^2}{2\theta T} f'(\xi(\theta)) - \frac{a^2}{2T\theta^2} f^2(\xi(\theta)) + O\left(\frac{1}{T^{3/2}}\right) \right) \right]^{\theta T}$$

$$\left[\exp\left(-\frac{a}{(1-\theta)\sqrt{T}} f(\xi(\theta)) - \frac{a^2}{T(1-\theta)} f'(\xi(\theta)) - \frac{a^2}{2T(1-\theta)^2} f^2(\xi(\theta)) + O\left(\frac{1}{T^{3/2}}\right) \right) \right]^{(1-\theta)T}$$

$$\approx \text{const.} \exp\left(-\frac{a^2}{2\theta} f^2(\xi(\theta)) - \frac{a^2}{2(1-\theta)} f^2(\xi(\theta)) + O\left(\frac{1}{\sqrt{T}}\right) \right)$$

$$= \text{const.} \exp\left(-\frac{a^2 f^2(\xi(\theta))}{2\theta(1-\theta)} \right)$$

$\Rightarrow \sqrt{T}\left(T(Y^T) - F^{-1}(\theta)\right)$ converges to normal with variance $\dfrac{\theta(1-\theta)}{f^2(\xi(\theta))}$